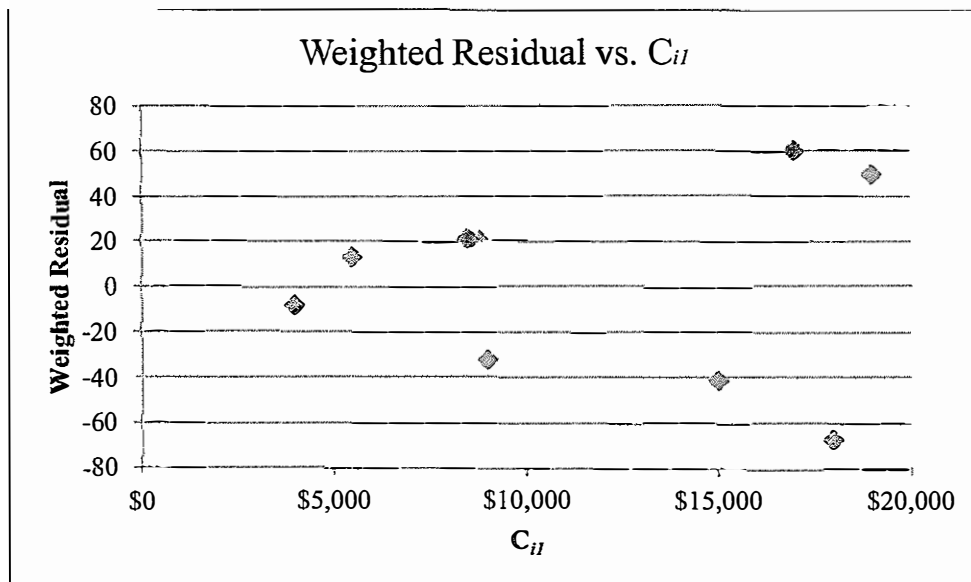
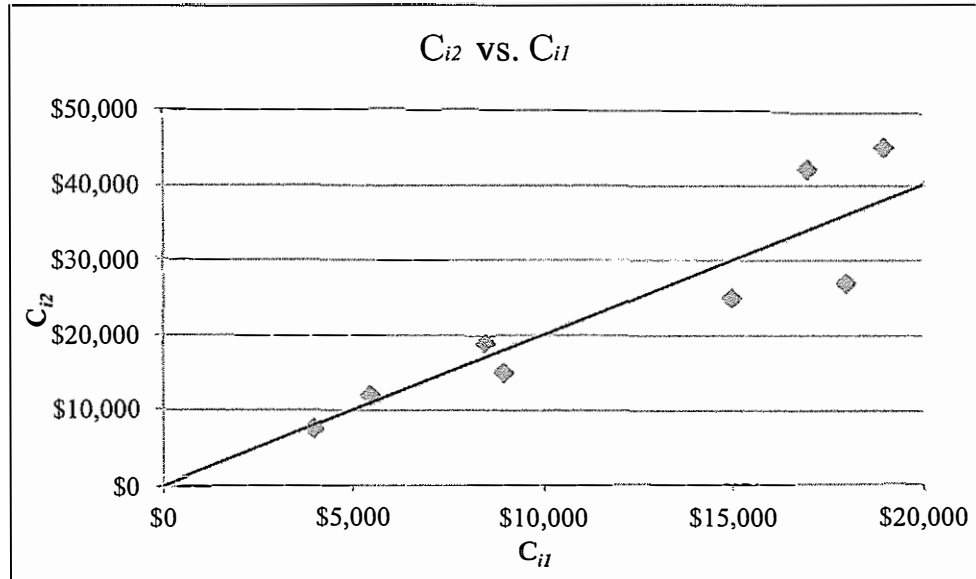


EXAM 7 – SPRING 2014

2. (1 point)

Given the following output from a company's reserving software:



- C_{i1} : Loss evaluated at 12 months for accident year i (\$000)
- C_{i2} : Loss evaluated at 24 months for accident year i (\$000)

Based on the two charts above, explain whether the chain-ladder method is appropriate for estimating ultimate loss.

CONTINUED ON NEXT PAGE

SAMPLE ANSWERS AND EXAMINER'S REPORT

QUESTION 2	
TOTAL POINT VALUE: 1	LEARNING OBJECTIVE: A2: Estimate parameters and unpaid claims using claims development models related to loss reserving methods such as: • Chain ladder • Cape Cod • Chain ladder plus calendar-year effects • Bornhuetter-Ferguson
SAMPLE ANSWERS	
<i>Sample Answer 1</i> The graph shows Ci_2 & Ci_1 has linear relationship and the line goes through the origin. It supports one of the CL assumptions; expected incremental loss is proportional to losses reported to date. Graph 2 shows that residuals are increasing as losses increase. This violates the variance assumption. Overall I think CL method is not appropriate. <i>Sample Answer 2</i> The first chart shows an estimate using CL approach (a linear fit with origin at zero). Overall this appears to be a pretty good fit, with the variance increasing in proportion to losses to date. This is also shown by the residuals. Overall, there is not a trend in the residuals up or down, but they do expand at the right tail. Since these are weighted residuals that could mean the proportion is not constant, but not enough evidence. Overall residuals are even around zero for any loss size, overall CL is a good fit. <i>Sample Answer 3</i> The graph of Ci_2 v Ci_1 is a straight line through the origin and appears to be a decent fit. This indicates that the CL method is appropriate. The second graph appears to show that the residuals are scattered around 0, however the variance appears to be larger as losses become larger. In order to determine if the CL method is appropriate for ultimate, we would need to see the graphs for other ages as well.	
EXAMINER'S REPORT	
The candidate was expected to understand the assumptions of the CL method and apply the knowledge to a simple example. In order to receive full credit the candidate was required to address both charts and demonstrate how they support or violate the Mack CL assumptions. Candidates that only stated the points fit the line well for the top chart lost credit. Candidates that rejected the approach due to larger errors for larger AY losses in top chart also lost credit. A common misinterpretation of the bottom chart was that candidates did not realize that the weighted residuals are adjusted for size of AY loss and that there should not be a pattern based on	

SAMPLE ANSWERS AND EXAMINER'S REPORT

size of loss if CL assumptions hold. About half the candidates received full credit and over 85% received at least .75 points.